

2026

Bright Coffee Shop Sales Analysis

DUE DATE: 31 August 2026

Introduction

You have been provided with a dataset titled 'Bright Coffee Shop Sales' which captures daily transactional information from a coffee shop. The business has recently appointed a new CDO whose mission is to grow the company's revenue and improve product performance. Your role as a Junior Data Analyst is to extract actionable insights from historical data and prepare comprehensive reports for their decision-making.

Objective

Use your statistics, SQL, and data visualisation skills to help Bright Coffee Shop understand:

- Which products generate the most revenue
- Which time of day the store performs best
- Sales trends across products and time intervals
- Recommendations for improving sales performance

Tools Allowed

You may use any tool at your disposal including but not limited to:

Project Planning	Coding Platforms	Data Visualisation	Presentation & Reporting
Miro	Microsoft SQL Server	Microsoft Excel	Microsoft PowerPoint
Figma	Deaktio	Power BI	Canva
	Google Fit	Tableau	Mix
	MS Word	Google Sheets	Figma

Tasks

Task 1: Planning & Architecture (Mini)

1.1. Develop data flow & architecture design for the solution.

- Where the data comes from (source)
- How it is processed (ETL pipeline)
- Where it is stored (destination)
- How it is analysed and presented

1.2. List the key insights that the team should deliver, including:

- Identify product category and time intervals
- High performing and low performing products
- Total revenue calculations

1.3. Outline the calculations to be performed:

- $\text{Total Amount} = \text{unit_price} * \text{transaction_qty}$
- Grouping by time interval (hourly/daily)
- Calculate total by product type and month

You can watch the video, this is just to give you an idea of what is expected from the final planning.

Task 2: Data Processing in Databricks

2.1. Convert the provided raw data to CSV

2.2. Load the CSV into Databricks

2.3. Perform data transformations:

- Create a new column 'transaction_time_bucket' to group transactions into 30-minute intervals (24 hours can be 96 time intervals)
- Calculate 'total_amount' property (some entries use comma as a decimal separator, e.g., '31' should be converted to '31.0')
- Calculate 'total_amount' = unit_price * transaction_qty
- Use SQL to group by product types, time buckets, etc.

You can watch the video, this is just to give you an idea of what is expected from the data processing.

Task 3: Data Analysis in Excel

3.1. Process the data in Databricks

3.2. Export the processed table to Microsoft Excel for your sales report

3.3. Create dashboards or pivot tables showing:

- Total revenue per product type
- Total time intervals for sales
- Quantity of items sold by product category
- Which selling product types are doing best
- Use charts and graphs to make the story visually appealing

Task 4: Presentation to the CEO

4.1. Highlight key insights from your analysis (qualitative insights) - Create a separate document for your presentation

2. Recommendations

- Marketing campaigns during low time slots
- Adjust more of the best-selling items
- Improve underperforming products

Next steps

- Document daily sales reporting
- Track sales performance across multiple locations in the future
- Implement loyalty programs based on peak customer time slots

Submission Guidelines (GitHub)

Submit the following items:

- Mini Plan Diagram
- Processed Data in CSV format (if you don't have it, provide the code to generate it)
- PowerPoint presentation
- SQL file with your code

